OS for Reverse Engineering Practice

DLL, PE format, TLS

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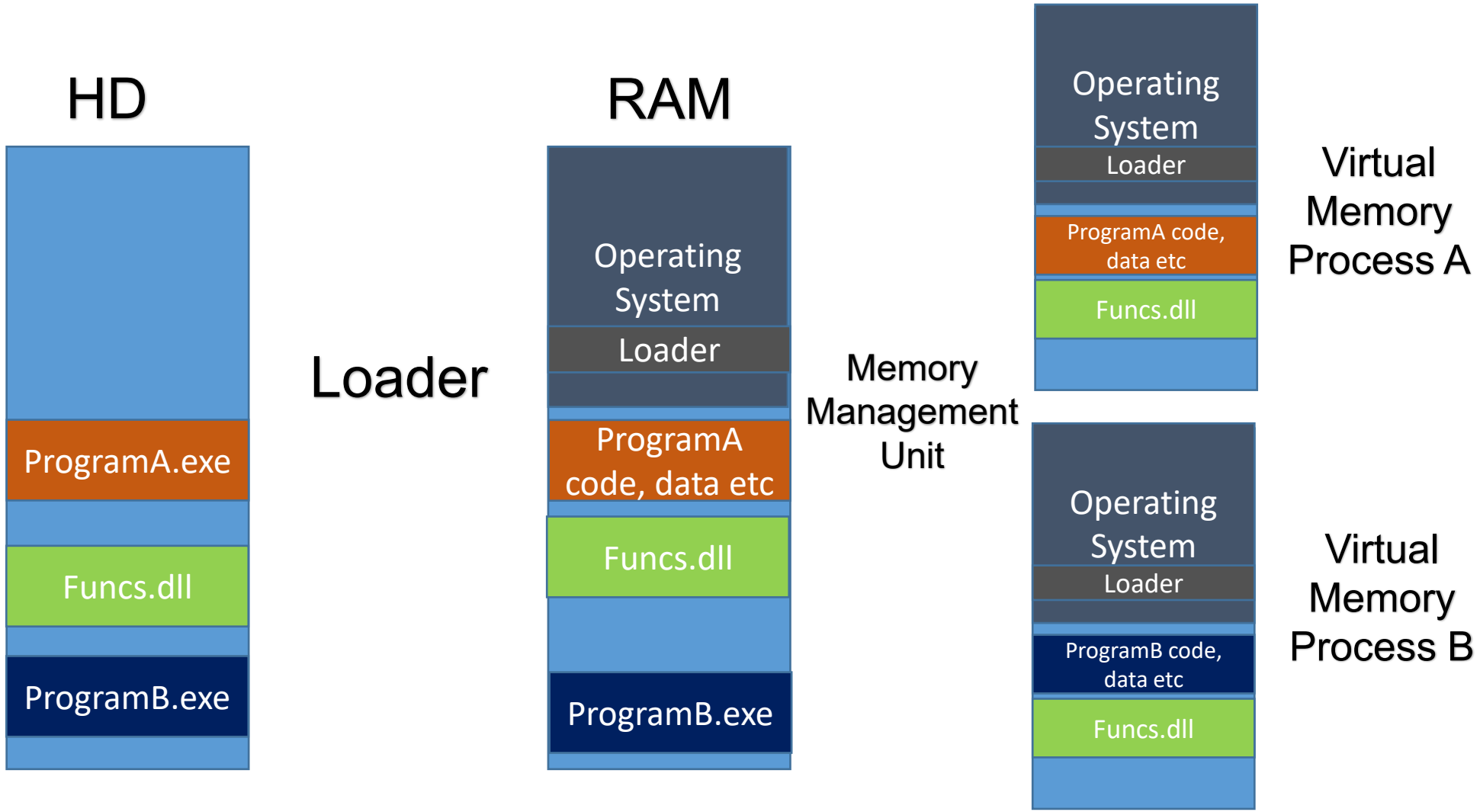
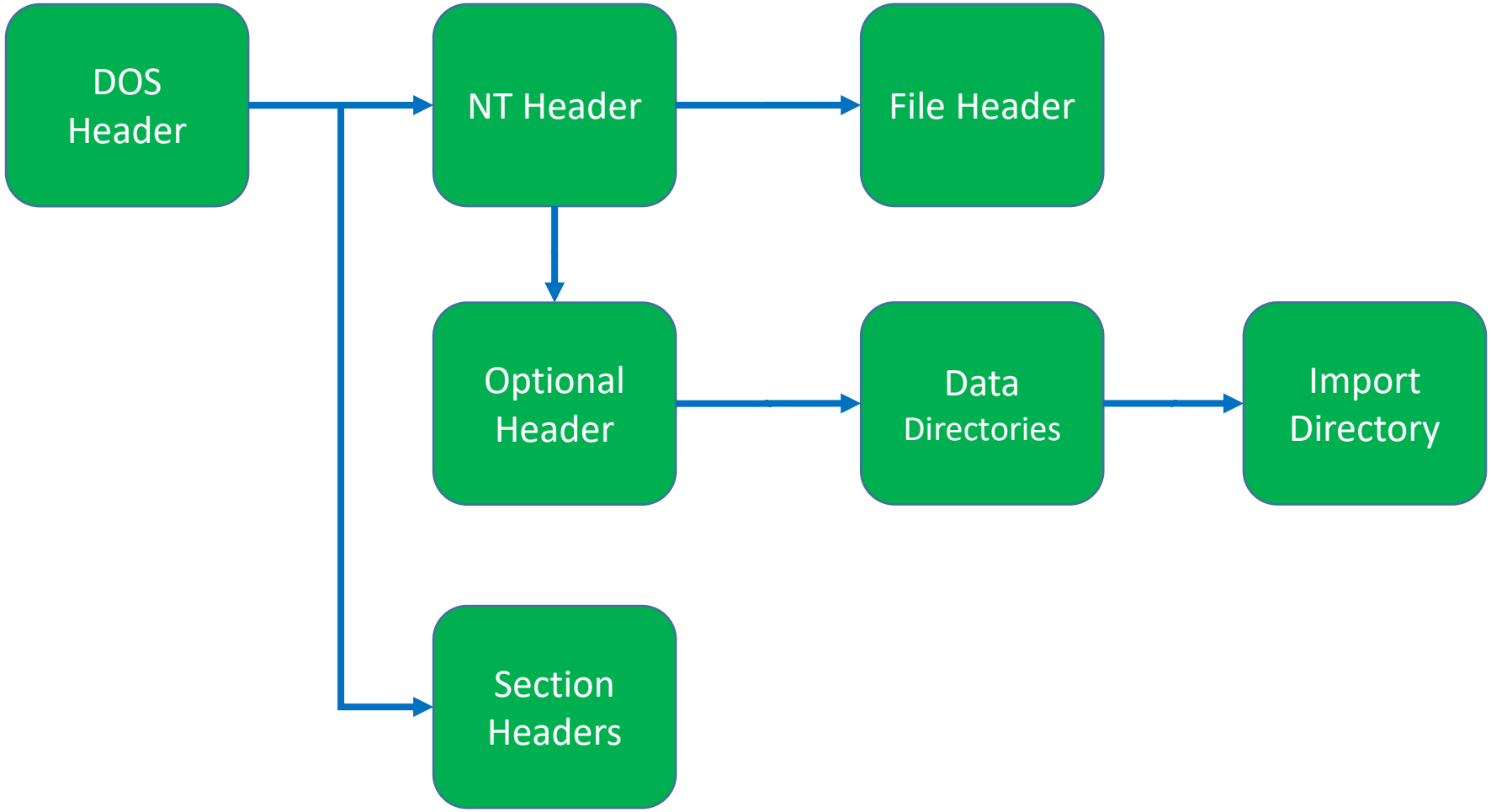
Topics

- What have we learned?
- DLL practice
- TLS theory and practice
- Format PE practice

From Disk to Virtual Memory



PE Format


DOS Header
NT Header
File Header
Optional Header
Data Directories
Import Directory
Section Headers

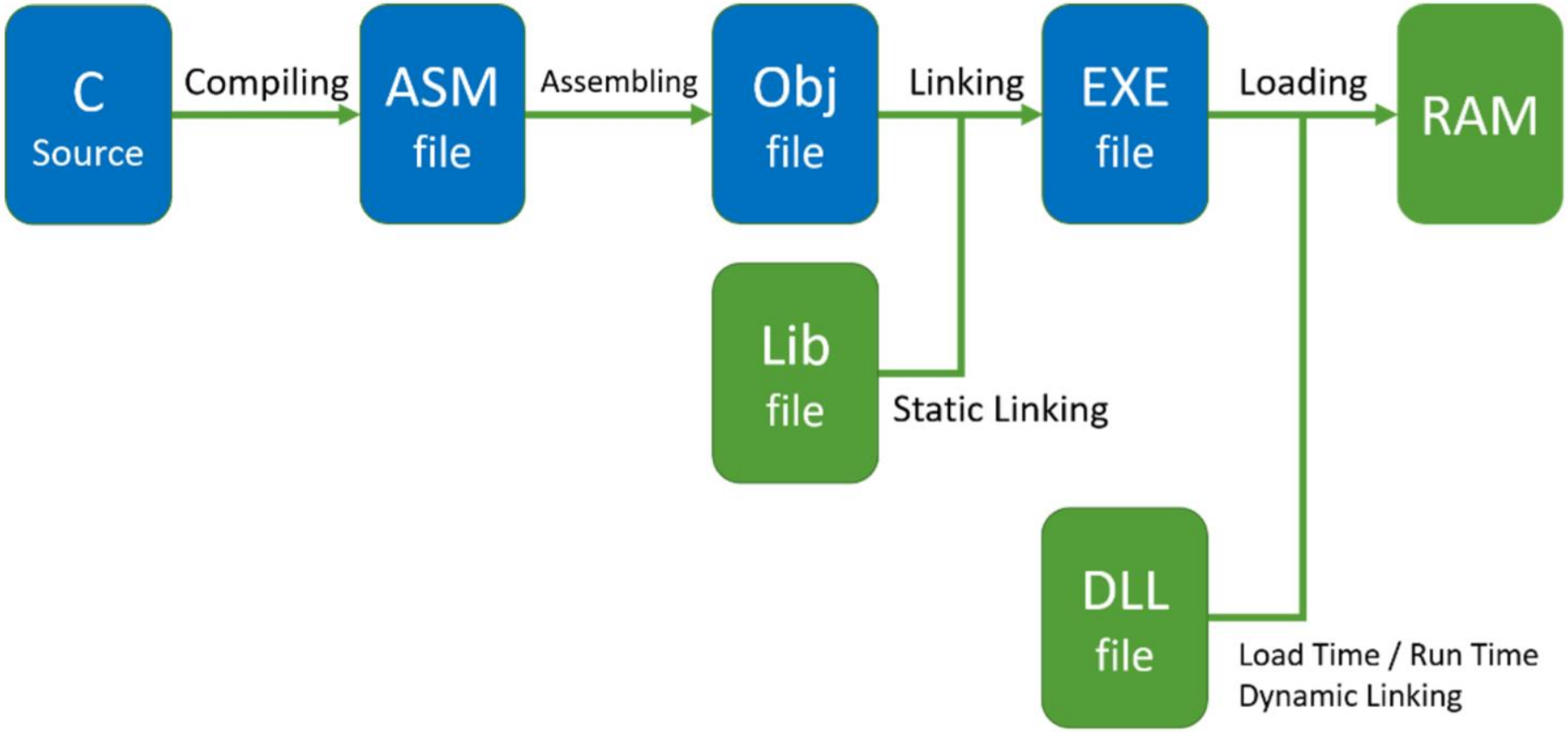
PE Format

- MZ, DOS stub
- Entry point
- Base address
- Memory sections
- DLL’s to load
- Relocation table
- Import address table
- DLL characteristics

DLL

- Dynamic Link Library
- A code which is shared among processes
  - No need for source code (intellectual property reserved)
  - Loaded to memory only once (saves memory)
- For example
  - user32.dll – Windows management functions for message handling, timers, menus, and communications
  - Kernel32.dll – Low-level operating system functions for memory management and resource handling

The complete process

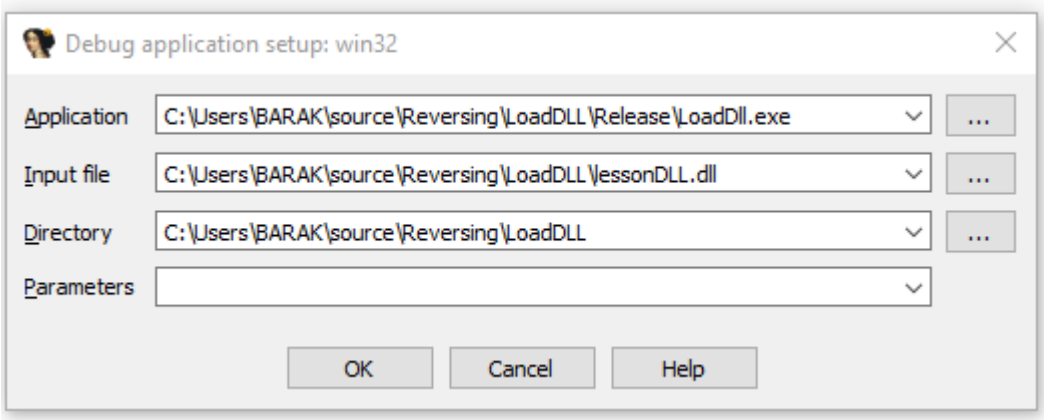


Fun with DLL

- https://data.cyber.org.il/reversing/game.zip
- Make it work!

IDA DLL Analysis

- Using CFF, find which functions are exported
- Use LoadLibrary + GetProcAddress to call them
- Configure IDA: Debugger -> Process Options



- https://data.cyber.org.il/reversing/lessonDLL.dll

TLS

- Thread Local Storage
- Can store data
- Can run functions ☺
- https://data.cyber.org.il/reversing/lessonTLS.exe
- Look at “exports” tab

Format PE Practice

- Find the flag!
- https://data.cyber.org.il/reversing/FindTheFlag.pdf